Scaling Relations and MRA

Course: Math 414

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Core Idea

At a high level, the process of processing (such as compressing or denoising) a signal $y = f(t)$ in practice is as follows:

1. **Sample**, we can use a power of two greater than Nyquist rate $2^J > 2v = \frac{\Omega}{\pi}$ for our purposes

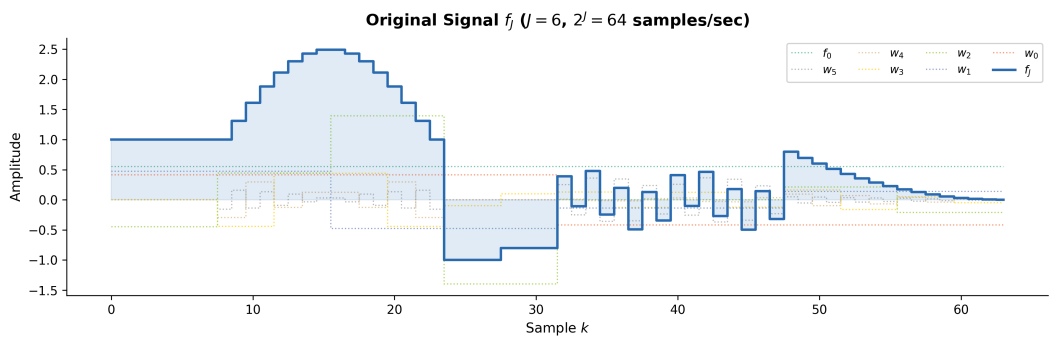

Figure 1: Sampling our function. Dotted lines are the decomposition as well

1. **Decompose** (with the algorithm below) f_J into a signal of the form $f_{J(x)} = \sum_{j=0}^{J-1} w_j + f_0$

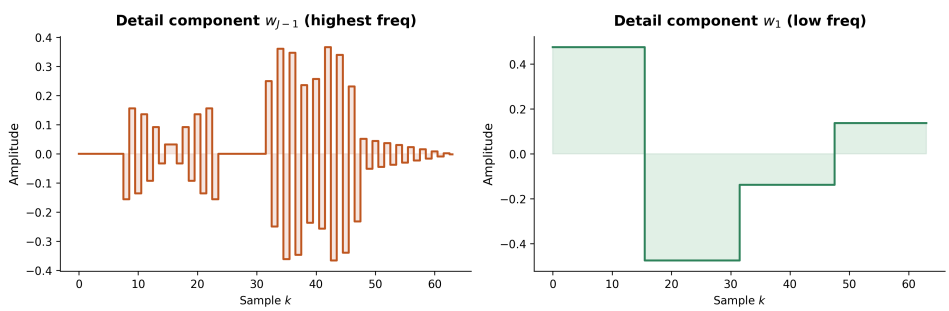

Figure 2: I just chose two random ones, the full decomposition is above

1. **Process** the signal.
   1. To filter out high frequencies, we can choose a max value of b_k^j. If we think about b_k as acting on ψ by increasing the height, then this means that those frequencies are gone
   2. To compress the signal, then b_k^j under a threshold (meaning they don't make a big difference), get thrown out

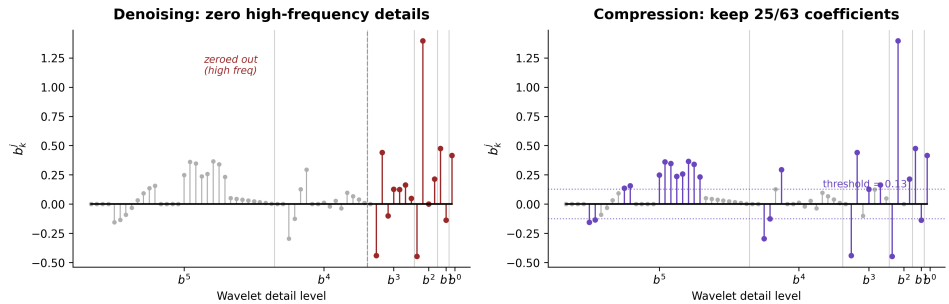

Figure 3: To remove high frequencies, we literally just remove the wavelets above a certain level. To compress, we remove wavelets which don't add much change

1. **Reconstruct** from our coefficients

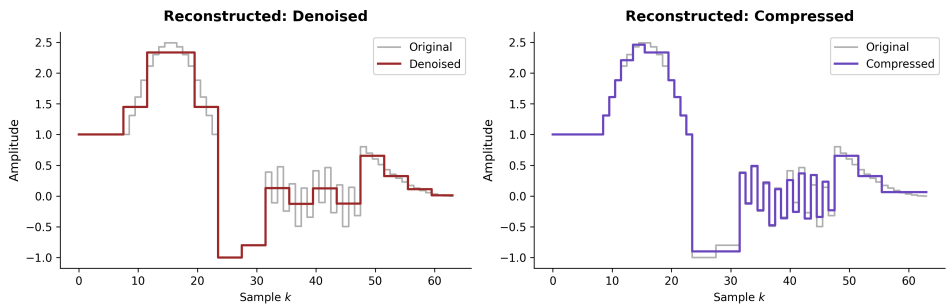

Figure 4: Just remaking the signal from our b_k that we got

Definitions

This notation is important

$$f_j \in V_j, f_j = \sum_{k \in \mathbb{Z}} a_k^j \underbrace{\phi(2^j x - k)}_{v_j} \tag{1}$$

Here is how we decompose our functions for the following proofs

$$\phi(x) = \phi(2x) + \phi(2x - 1) \tag{2}$$

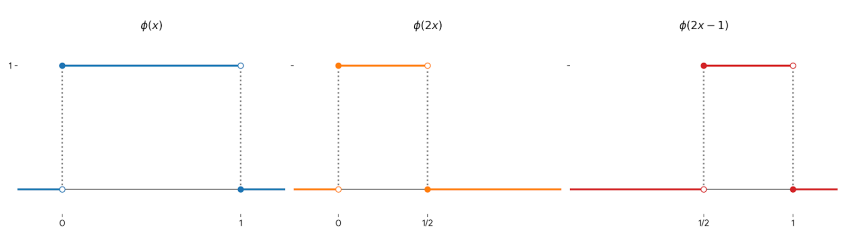

Figure 5: How we can express the haar scaling function

$$\psi(x) = \phi(2x) - \phi(2x - 1) \tag{3}$$

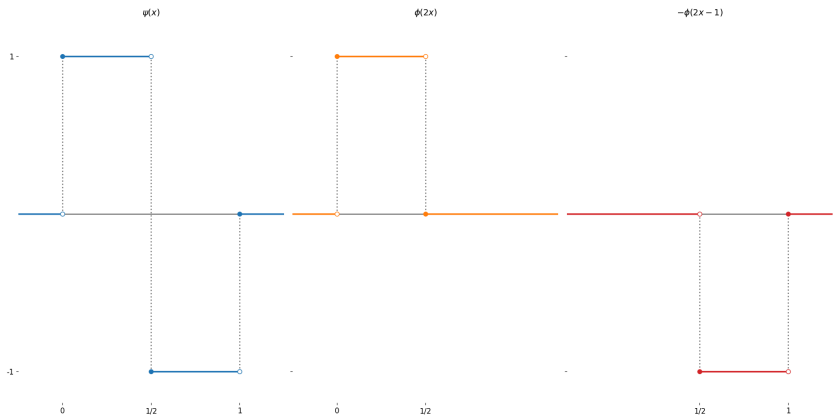

Figure 6: How we can express the haar wavelet function

If we add them

$$\phi(2x) = \frac{1}{2}(\phi(x) + \psi(x)) \tag{4}$$

If we subtract them

$$\phi(2x-1) = \frac{1}{2}(\phi(x) - \psi(x)) \tag{5}$$

If we replace x with $x-k$

$$\phi(x-k) = \phi(2x-2k) + \phi(2x-2k-1) \tag{6}$$

$$\psi(x-k) = \phi(2x-2k) - \phi(2x-2k-1) \tag{7}$$

$$\phi(2x-2k) = \frac{1}{2}(\phi(x-k) + \psi(x-k)) \tag{8}$$

$$\phi(2x-2k-1) = \frac{1}{2}(\phi(x-k) - \psi(x-k)) \tag{9}$$

If we replace x with $2^{j-1}x$ in Equation 4, Equation 5

$$\phi(2^j x) = \frac{1}{2}(\phi(2^{j-1}x) + \psi(2^{j-1}x)) \tag{10}$$

$$\phi(2^j x - 1) = \frac{1}{2}(\phi(2^{j-1}x) - \psi(2^{j-1}x)) \tag{11}$$

Haar Decomposition Formula

$$a_k^0 = \frac{1}{2}(a_{2k}^1 + a_{2k+1}^1) \tag{12}$$

$$b_k^0 = \frac{1}{2}(a_{2k}^1 - a_{2k+1}^1) \tag{13}$$

More generally

$$f_j = w_{j-1} + f_{j-1} \tag{14}$$

$$w_{j-1} = \sum_{k\in\mathbb{Z}} b_k^{j-1}\psi(2^{j-1}x - k) \in W_{j-1} \tag{15}$$

$$f_{j-1} = \sum_{k\in\mathbb{Z}} a_k^{j-1}\psi(2^{j-1}x - k) \in V_{j-1} \tag{16}$$

$$a_k^{j-1} = \frac{1}{2}\left(a_{2k}^j + a_{2k+1}^j\right) \tag{17}$$

$$b_k^{j-1} = \frac{1}{2}\left(a_{2k}^j - a_{2k+1}^j\right) \tag{18}$$

This gives us

$$f_j = w_{j-1} + w_{j-2} + ... + w_0 + f_0 \tag{19}$$

Haar Reconstruction Formula

$$(a_{2k}^1 = a_k^0 + b_k^0 \tag{20}$$

$$(a_{2k+1}^1 = a_k^0 - b_k^0 \tag{21}$$

$$f_J = \sum_{k\in Z} a_k^j \phi(2^J x - k) \tag{22}$$

Proof for decomposition

Let's suppose we have $f_1 \in V_1$, with O.N. basis $\{\phi(2x-k), k \in \mathbb{Z}\}$

$$f_1(x) = \sum_{k\in\mathbb{Z}} a_k^1 \phi(2x - k) \tag{23}$$

Split into even and odd

$$f_1(x) = \sum_{k\in\mathbb{Z}} a_{2k}^1 \phi(2x - 2k) + \sum_{k\in\mathbb{Z}} a_{2k+1}^1 \phi(2x - 2k - 1) \tag{24}$$

From Equation 8 and Equation 9, substitute them

$$f_1(x) = \sum_{k\in\mathbb{Z}} a^1_{2k}\left(\frac{1}{2}(\phi(x-k)+\psi(x-k))\right) + \sum_{k\in\mathbb{Z}} a^1_{2k+1}\left(\frac{1}{2}(\phi(x-k)-\psi(x-k))\right) \tag{25}$$

Collecting terms we have

$$f_1(x) = \sum_{k\in\mathbb{Z}}\left(\frac{1}{2}(a^1_{2k}+a^1_{2k+1})\phi(x-k) + \frac{1}{2}(a^1_{2k}-a^1_{2k+1})\psi(x-k)\right) \tag{26}$$

$$f_1(x) = \sum_{k\in\mathbb{Z}}\frac{1}{2}(a^1_{2k}+a^1_{2k+1})\phi(x-k) + \sum_{k\in\mathbb{Z}}\frac{1}{2}(a^1_{2k}-a^1_{2k+1})\psi(x-k) \tag{27}$$

But we also know from before that $f_1 = f_0 + w_0$

$$f_1(x) = \sum_{k\in\mathbb{Z}} a^0_k\phi(x-k)\sum_{k\in\mathbb{Z}} b^0_k\psi(x-k) \tag{28}$$

The coefficients from Equation 27 and Equation 28 are the same, and we get to the decomposition formula above.